

Synchronized Determinant Prefixes and a Two-Order Orbit–Head Compensation Obstruction

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Abstract

We put the weighted prefix–tail determinant criterion of RH-288 [4] on the physical first–alias clock. If $L = \log(1/\sigma)$ and $k = L/(2 \log \lambda) + O(1)$, the integer cut $u = 4k$ is still inside the one–alias window. Reapplying the RH-282 mass–and–cap estimate at this cut, rather than identifying it with RH-282’s displayed $\lceil 4L \rceil$ cut, proves that both the noisy modulus–complement tail and the deterministic target tail vanish on the disk $|z| \leq R$, $R = 7/5$.

On one common Hardy normalization define the direct complement prefix P_u , the full–trace prefix E_u , and the head/counterloop prefix D_u . The exact identity $p_n = q_n - d_n$ gives the sharp finite–sum inequality $|P_u - E_u| \leq D_u$. Thus, if the head budget vanishes on the same clock, direct complement closure is equivalent to full–trace closure; the RH-330 critical extraction then reduces the remaining condition to the same–clock triple $(D_u, E_{\text{off}}, q_{2k}/H_k)$.

The critical orbit atom of RH-338 and the mandatory lower sideband atom of RH-339 force two signed compensation equations at orders $2k$ and $2k - 2$. Any proof that takes separate absolute values of the orbit, diffuse, and head pieces has a divergent two–atom majorant. This is a strict synchronization theorem and a scoped obstruction to a cancellation–blind route. It is not a lower bound for the fully signed prefix: the aggregate signed complements, the head budget, and E_{off} remain NOT.TESTABLE. RH-288 is not activated, Gates A–E remain false/open, and no Hilbert–Polya or Riemann–hypothesis conclusion follows.

1 The physical cut and its two tails

Let the physical first–alias clock be the RH-337 clock [2]:

$$L = \log(1/\sigma), \quad k = k_\sigma \in \mathbb{N}, \quad k \geq 2, \quad k = \frac{L}{2 \log \lambda} + O(1), \quad R = \frac{7}{5}. \quad (1)$$

The common determinant cut is

$$u = 4k, \quad 2k < u \leq 4k. \quad (2)$$

The physical constants use $r_H = 17/20$, $q = 1/2$, and

$$q_* = (r_H \lambda)^{-1}, \quad qR = \frac{7}{10}, \quad q_* R = \frac{28}{17\lambda} < 1. \quad (3)$$

The last strict inequality is the exact RH-262 boundary certificate [1]; RH-336 supplies the certified bound $\lambda < 17/10$ [6].

Write $\tau_{\sigma,n}$ for the modulus–complement trace in the RH-288 normalization and $a_n = a_n^{\text{num}}$ for the deterministic numerator anchor. The direct complement tail and target tail at the cut are

$$S_{\sigma,u}(R) = \sum_{n \geq u} \frac{|\tau_{\sigma,n}| R^n}{n}, \quad T_u(R) = \sum_{n \geq u} \frac{|a_n| R^n}{n}. \quad (4)$$

Theorem 1.1 (Short-clock tail synchronization). *Under the RH-282 mass-and-cap estimate [9] and the RH-267/RH-268 deterministic all-order envelope [7, 11],*

$$S_{\sigma,4k}(R) \longrightarrow 0, \quad T_{4k}(R) \longrightarrow 0 \quad (\sigma \downarrow 0). \quad (5)$$

Proof. RH-282 gives, for the modulus-complement sequence and every integer $n \geq 2$,

$$|\tau_{\sigma,n}| \leq C \sigma^{-1} q^{n-2} \quad (6)$$

with a fixed harmless constant C . Hence

$$S_{\sigma,4k}(R) \leq \frac{C q^{-2}}{1 - qR} \frac{\sigma^{-1} (qR)^{4k}}{4k}. \quad (7)$$

Since $k = L/(2 \log \lambda) + O(1)$,

$$\sigma^{-1} (qR)^{4k} = O(\sigma^{\delta_N}), \quad \delta_N = \frac{2 \log(10/7)}{\log \lambda} - 1 > 0. \quad (8)$$

The strict sign follows from $\lambda < 17/10 < (10/7)^2$. This proves the first limit.

RH-267 gives $|a_n| < 48 q_*^n$ for every $n \geq 2$. Therefore

$$T_{4k}(R) \leq \frac{48 (q_* R)^{4k}}{4k(1 - q_* R)}, \quad (9)$$

which tends to zero by (3). The argument is a direct reapplication of the exact RH-282 mass bound at $4k$; it does not claim that the displayed RH-282 choice $\lceil 4L \rceil$ is equal to $4k$. $\square$

Remark 1.2. The tail theorem closes only the two analytic tail leaves. It does not identify a physical head, a counterloop, or a signed prefix, and therefore it does not activate RH-288.

2 A sharp same-clock prefix identity

Let $q_{\sigma,k,n}$ be the Hardy full-trace constituent of RH-334 [5] and let $d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}$ be its noisy-head/counterloop defect. RH-334 gives the exact identity, for every finite order in the common data type,

$$p_{\sigma,k,n} := \tau_{\sigma,n} - a_n = q_{\sigma,k,n} - d_{\sigma,k,n}. \quad (10)$$

Define the three nonnegative finite budgets

$$P_u(R) = \sum_{2 \leq n < u} \frac{|p_{\sigma,k,n}| R^n}{n}, \quad E_u(R) = \sum_{2 \leq n < u} \frac{|q_{\sigma,k,n}| R^n}{n}, \quad D_u(R) = \sum_{2 \leq n < u} \frac{|d_{\sigma,k,n}| R^n}{n}. \quad (11)$$

Theorem 2.1 (Sharp prefix synchronization). *For every finite common cut u and radius R ,*

$$|P_u(R) - E_u(R)| \leq D_u(R). \quad (12)$$

Consequently, on any common moving clock,

$$D_u(R) \rightarrow 0 \implies (P_u(R) \rightarrow 0 \iff E_u(R) \rightarrow 0). \quad (13)$$

Proof. For each order, (10) and the reverse triangle inequality give

$$||p_{\sigma,k,n}| - |q_{\sigma,k,n}|| \leq |d_{\sigma,k,n}|.$$

Multiply by R^n/n and sum over the finite set $2 \leq n < u$. This is (12). If $D_u \rightarrow 0$, the difference of the two nonnegative budgets tends to zero, proving (13). $\square$

For the one-alias cut (2), RH-330 isolates the critical order [10]:

$$E_u(R) = E_{\text{off},u}(R) + \frac{|q_{\sigma,k,2k}|}{2H_k}, \quad H_k = kR^{-2k}, \quad u = 4k. \quad (14)$$

Combining (13) and (14) gives the exact conditional three-budget criterion

$$D_u \rightarrow 0 \text{ and } P_u \rightarrow 0 \iff D_u \rightarrow 0, \quad E_{\text{off},u} \rightarrow 0, \quad q_{\sigma,k,2k} = o(H_k). \quad (15)$$

This is a same-clock statement; it permits no splice of a head estimate, critical estimate, and off-alias estimate taken on different clocks.

Corollary 2.2 (Conditional determinant reduction). *At $u = 4k$, the two tails in Theorem 1.1 vanish. Hence the RH-288 determinant quotient would converge uniformly on $|z| \leq R$ if the left-hand side of (15) were proved. The repository does not prove those prefix budgets, so RH-288 remains inactive.*

3 Two mandatory orbit atoms

The critical boundary orbit of RH-338 gives a signed far subledger [3]

$$\mathcal{R}_{k,\text{orb}} = -D_k^{\text{orb}}, \quad \frac{D_k^{\text{orb}}}{H_k} \rightarrow +\infty, \quad D_k^{\text{orb}}/H_k = \frac{2}{C_M}(\beta R)^{2k}(1 + o(1)), \quad (16)$$

where $\beta R > 1$ is the exact RH-336 certificate [3, 6]. Put

$$C_k^0 := q_{\sigma,k,2k} + D_k^{\text{orb}}. \quad (17)$$

The mandatory lower sideband of RH-339 has $m = k - 1$ [8] and

$$\mathcal{R}_{k,-,\text{orb}} = -D_m^{\text{orb}}, \quad \frac{D_m^{\text{orb}}}{H_m} = \frac{2}{C_M}(\beta R)^{2m}(1 + o(1)) \rightarrow +\infty, \quad H_m = mR^{-2m}. \quad (18)$$

Put

$$C_k^- := q_{\sigma,k,2m} + D_m^{\text{orb}}. \quad (19)$$

Both $2k$ and $2m = 2k - 2$ occur in the prefix $2 \leq n < u$.

Theorem 3.1 (Adjacent-order orbit–head compensation). *If the direct complement prefix satisfies $P_{4k}(R) \rightarrow 0$, then*

$$C_k^0 - d_{\sigma,k,2k} = D_k^{\text{orb}} + o(H_k), \quad (20)$$

$$C_k^- - d_{\sigma,k,2m} = D_m^{\text{orb}} + o(H_m). \quad (21)$$

The required relative precisions are respectively

$$o\left((\beta R)^{-2k}\right), \quad o\left((\beta R)^{-2(k-1)}\right). \quad (22)$$

If in addition $D_{4k}(R) \rightarrow 0$, the head terms in (20)–(21) are themselves $o(H_k)$ and $o(H_m)$, so both diffuse complements must pay their orbit atoms internally.

Proof. By (10), (17), and (19),

$$p_{\sigma,k,2k} = -D_k^{\text{orb}} + C_k^0 - d_{\sigma,k,2k},$$

$$p_{\sigma,k,2m} = -D_m^{\text{orb}} + C_k^- - d_{\sigma,k,2m}.$$

Since both orders occur in the nonnegative prefix,

$$P_{4k}(R) \geq \frac{|p_{\sigma,k,2k}|}{2H_k} + \frac{|p_{\sigma,k,2m}|}{2H_m}. \quad (23)$$

If the left side tends to zero, each numerator is respectively $o(H_k)$ and $o(H_m)$, which gives (20) and (21). Division by the asymptotics in (16) and (18) gives (22). Finally, $D_{4k} \rightarrow 0$ implies each displayed head summand is $o(H_k)$ or $o(H_m)$ by nonnegativity of D_{4k} . $\square$

Proposition 3.2 (Scoped failure of separate absolute majorants). *Any proof that takes separate absolute values of the two orbit atoms before their signed diffuse/head complements contains the atom submajorant*

$$\frac{D_k^{\text{orb}}}{2H_k} + \frac{D_m^{\text{orb}}}{2H_m} = \frac{1}{C_M} \left((\beta R)^{2k} + (\beta R)^{2k-2} \right) (1 + o(1)) \longrightarrow +\infty. \quad (24)$$

This proposition does not lower-bound P_{4k} or $E_{\text{off},4k}$: a fully signed complement may still cancel both atoms.

Proof. Insert (16) and (18) into the two weighted atom terms. Since $\beta R > 1$, the sum diverges. The final scope statement follows because the complements in (17) and (19) retain their signs. $\square$

4 Executable audit and claim boundary

The artifact evaluates the exact cut indices and the two positive tail exponents at $k = 3, 5, 9, 17, 33$. It also reports the diagnostic two-atom majorant using the printed C_M value; that decimal is not an interval certificate and is not used for a theorem sign. The tests lock the one-alias window, the $2k$ and $2k - 2$ orders, the deterministic result, and the claim firewall.

The maximum justified conclusion is therefore:

- the analytic tails can be synchronized at $u = 4k$ by direct corollaries of existing bounds;
- same-clock head closure makes direct and full-trace prefix closure equivalent, with the exact critical/off-alias extraction;
- the two physical orbit atoms impose the necessary signed laws above, and cancellation-blind absolute splitting is impossible;
- no moving-order signed complement or head estimate is available, so aggregate prefix closure and nonclosure are both `NOT.TESTABLE`.

No physical determinant gluing is activated. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace or completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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